Junbin Chen

LLM Evaluation · Statistical Analysis · Python · PhD Statistics

2302, 222 Marsh Wall, London E14 9PH

07521 356683 | londonjunbinchen@gmail.com | Right to work in the UK

PROFILE

Statistician who has spent five years finding out whether models can be trusted and showing the evidence. At Lloyds Banking Group I led the independent review of large language model use cases in Markets: deciding what to test, building evaluation sets and consistency checks across repeated runs, hunting for the quiet failures (hallucination, inconsistency, data leakage, misuse) and writing it up so that both quants and a risk committee could act on it. Before that, four years validating pricing and algorithmic trading models by re-implementing them, and building surveillance analytics that pull anomalous behaviour out of large, noisy trade datasets in Python. PhD in statistics on predictive inference. Daily AI power user, including an agentic application I built on Claude this year. My Python is analytical rather than production-grade; I am building that with an Inspect-based evaluation project now. I want to apply the same rigour to frontier models before they ship.

CORE COMPETENCIES

LLM evaluation	Designing evaluations for LLM systems: test-set construction, behavioural and consistency testing across repeated runs, failure-mode taxonomies, red-team style probing for misuse and data leakage, rubric-based and automated scoring, reproducible Python test harnesses; building evals in Inspect (in progress)
Data analysis & anomaly detection	Extracting signal from large, messy datasets: surveillance analytics on trade and order data, risk-factor history reconstruction, back-testing against known outcomes; statistical inference, hypothesis testing, time-series analysis; knowing when a number is not the answer and qualitative review is
Model scrutiny	Four years of independent model validation: re-implementing models to find where they diverge, challenging assumptions and implementation, documenting limitations; validation of algorithmic trading models for unintended behaviour and conduct risk
Writing & communication	Validation reports and committee papers that quants and non-technical executives both acted on; peer-reviewed publication and conference talks; three years of university teaching; fluent English, native Mandarin
Technical & AI tooling	Python (primary; pandas, numpy), SQL, R, Java, C++; daily use of Claude and other frontier models for analysis and coding, with judgement about which model for which task; Anthropic API and agentic patterns; reading on post-training (RLHF, SFT, reasoning training) and scheming/alignment-faking research
Qualifications	PhD Statistics (Durham), FRM Part 1

PROFESSIONAL EXPERIENCE

Career break Feb 2026 – present

Planned break after five continuous years at Lloyds, used to build an agentic application on Claude, to study LLM evaluation practice, post-training and AI-safety research on scheming and alignment faking, and to build an Inspect-based behavioural evaluation suite (GitHub link on request).

Lloyds Banking Group, London Oct 2020 – Jan 2026

Manager, Market Risk AI team Jun 2025 – Jan 2026

- Led the independent review of generative AI use cases in Markets with no existing method: defined what to test, built evaluation sets and consistency checks across repeated runs, probed for hallucination, data leakage and misuse, classified failure modes, set the controls and conditions of approval, and wrote it up for the Model Risk Committee.

- Built trade surveillance analytics in Python on client transaction data with Compliance and the Front Office: designed the detection logic, tuned thresholds, back-tested on historical trades to measure true and false positives, and documented the control to audit standard.

Manager, Market Risk Approval Apr 2024 – Jun 2025

- Delivered FRTB analysis in Python on the trading book, covering risk-factor eligibility, non-modellable risk factors and capital impact, and explained the results to Risk management.
- Owned rate backfilling and VaR remediation for the IBOR transition (Python, C++, VBA): rebuilt curve histories for transitioning benchmarks and reconciled VaR across the affected book to a regulatory deadline.
- Reviewed and approved changes to the bank's market risk models: challenged model owners on methodology, data and testing, wrote recommendations for the Model Risk Committee and answered questions from senior management.
- Ran ongoing performance monitoring of risk models with the methodology team, identifying degradation, raising and closing findings, and contributing to the validation methodology and annual review cycle.

Assistant Manager, Pricing Model Validation Oct 2020 – Apr 2024

- Independently validated the money-market pricer and its risk sensitivities: reviewed documentation and data, rebuilt the model in Python, benchmarked against production, assessed limitations and wrote a recommendation.
- Validated the e-rates pricing engine (Java) against its documented methodology, including review of production code and testing.
- Validated the bank's eFX algorithmic trading models (Java): design, performance monitoring, stability and conduct-risk controls.
- Authored validation reports to governance and audit standards and presented conclusions and conditions of approval to model owners and the Model Risk Committee; tracked findings to closure.

United Credit Ratings Co. Ltd, Beijing — Intern May – Jul 2015

- Migrated corporate bond rating reports and issuer financial statements to a new system and carried out due diligence on data accuracy.

EDUCATION AND QUALIFICATIONS

PhD Mathematics (Statistics), Durham University 2019

Thesis: *Contributions to Nonparametric Predictive Inference for Bernoulli Data with Applications in Finance* (imprecise probability, asset and option pricing).

MSc Mathematical Sciences, Durham University 2014

BA Business Administration, China Agricultural University / University of Bedfordshire 2012

FRM Part 1, GARP

While at Durham: undergraduate tutor and teaching assistant for 12 tutorial groups, and elected research-student representative to the departmental board of studies.

PUBLICATIONS AND PRESENTATIONS

- Chen, J. et al., "On nonparametric predictive inference for asset and European option trading in the binomial tree model", *Journal of the Operational Research Society*, 2019.
- Presented at the Financial Engineering and Banking Society conference (Glasgow, 2017) and the Winter School on Mathematical Finance (Lunteren, 2018).

ADDITIONAL INFORMATION

- Languages: Chinese (native), English (fluent), French (lower-intermediate), Spanish (beginner).
- Interests: distance running (Hampton Court and Paris half marathons), public science outreach at Durham's Celebrate Science festival.
- References available on request.